# Digital Forensics Investigation

Simple Step-by-Step Project Explanation & Viva Voce Guide

## 1. What is this Project in Simple Words?

Think of this project like a detective investigation, but for computers. An employee at a company is suspected of stealing confidential files using a USB drive. Before getting caught, the employee tried to hide what they did by:
1. Renaming a secret blueprint photo (.jpg) into a text file (.txt) so it looks innocent.
2. Deleting a confidential client list so nobody would see it.

Our job as digital forensic analysts is to investigate the USB drive, prove scientifically that the evidence was not touched or changed, uncover the renamed photo, recover the deleted client list, and prepare a proper report that can be shown to a judge in court.

## 2. The Three Golden Rules Every Examiner Must Follow

**Rule 1: Acquisition (Never touch the original).** If you plug a USB into a normal Windows PC, Windows immediately changes file dates and writes hidden files. This ruins the evidence! Instead, we use a Write-Blocker to make an exact sector-by-sector clone called a disk image (.dd), and we do all our tests on that copy.

**Rule 2: Integrity (Prove nothing was changed).** We use mathematical algorithms called hash functions (like MD5 and SHA-256) to calculate a unique 'digital fingerprint' of the drive. If even a single letter or bit changes later, the fingerprint completely breaks. This proves the evidence is authentic.

**Rule 3: Chain of Custody (Keep a signed paper trail).** We keep a formal table listing who seized the USB, who created the copy, what time it happened, and who received it. Without this paper trail, no court will accept the evidence.

## 3. Step-by-Step Breakdown: What We Actually Did

### Step 1: Created the Evidence Media (suspect_device.dd)

Why couldn't we use a normal Windows folder? Because a normal folder doesn't have disk sectors, partition boot records, or unallocated space. Forensic tools like FTK Imager and Autopsy inspect the entire file system structure. So we created a virtual disk image formatted with FAT12 (1.41 MB) containing normal files, a deleted file, a renamed file, and hidden notes.

### Step 2: Bit-Stream Physical Acquisition (FTK Imager)

We used FTK Imager in read-only write-blocked mode. We selected 'Physical Drive' to copy every single sector from start to finish (all 2,880 sectors). FTK Imager computed the baseline hashes and confirmed a 100% clean copy without errors.

### Step 3: Verified Integrity with Python (forensic_hasher.py)

We wrote a Python script using the hashlib library. It read the raw image in 64KB chunks and calculated:

- MD5: 8b71c348840eb453e0314961737ac26a
- SHA-1: cf4850eb15729162732448cbeeaf7e17acb9475f
- SHA-256: 3b5a7762f2c49c891da888c8be90f5c67baa4fdecda9c2764d94a8845e75a01c

Our Python hashes matched the FTK Imager baseline 1:1, proving the image was completely intact.

### Step 4: Demonstrated Tamper Detection (The Avalanche Effect)

We took a duplicate copy and flipped just 1 single bit at byte 1024. When we re-ran the hash, the SHA-256 changed completely: 127 of the 256 bits changed (~49.6%). This proves to a teacher or judge that even a tiny modification is immediately caught.

### Step 5: Hex Analysis in HxD: Unmasked the Renamed File

The suspect renamed a secret photo to MEMO.TXT. But in HxD Hex Editor, we looked at the first 4 bytes of the file: FF D8 FF E0. In computer science, this is the 'magic number' of a JPEG photo! We also found the comment 'EXFILTRATED_SYSTEM_DIAGRAM'. This proved the file was secretly an exfiltrated diagram, not text.

### Step 6: Recovered the Deleted Client List (hex_analyzer.py)

When you delete a file in a FAT file system, the computer does NOT erase the file. It only changes the first letter in the directory table to hex 0xE5. The actual data clusters remain on disk. Our script read the directory table, found the 0xE5 flag, traced the starting cluster (Cluster 5, offset 0x4800), and carved the full 200 bytes of RECOVERED_CLIENTS.CSV back to life!

### Step 7: Recorded the Chain of Custody

We recorded every transfer in a formal table with dates, times, names, and SHA-256 hashes so the evidence is legally admissible under Section 65B of the Evidence Act.

## 4. Top 15 Viva Voce Questions & Simple Answers

*Here are the most common questions teachers ask, with simple, direct answers you can say out loud:*

**Q1: What is Digital Forensics in simple words?**

**Answer:** It is the scientific process of collecting, verifying, and analyzing computer evidence so that it can be presented in a court of law without being altered.

**Q2: Why should you never investigate the original evidence directly?**

**Answer:** Because operating systems like Windows automatically change access dates, write temporary files, and modify data as soon as you plug a drive in. That destroys the evidence.

**Q3: What is the difference between a Physical Image and a Logical Image?**

**Answer:** A logical image only copies currently visible files. A physical bit-stream image copies every single raw sector from start to finish, including deleted files, slack space, and unallocated space.

**Q4: What is a Write-Blocker?**

**Answer:** It is a hardware device or software driver that blocks all write commands while allowing read commands. It makes sure the drive is 100% read-only so nothing can be changed during imaging.

**Q5: What is a Hash Function?**

**Answer:** A hash function is a one-way mathematical formula that produces a unique digital fingerprint of a file (like SHA-256). If the file changes by even one bit, the hash changes completely.

**Q6: Why did you use both MD5 and SHA-256?**

**Answer:** MD5 is fast and widely supported in older legal cases, but SHA-256 is much stronger and collision-resistant. Using both gives full backward compatibility and modern security.

**Q7: What is the Avalanche Effect?**

**Answer:** It means if you change just 1 bit in the input data, about 50% of the output hash digest changes completely. We demonstrated this by flipping 1 bit, and 127 out of 256 bits changed (~49.6%).

**Q8: What is a Magic Number (File Signature)?**

**Answer:** It is the unique hexadecimal bytes placed at the very beginning of a file that identify what type of file it really is, regardless of its extension.

**Q9: What are common magic numbers to know?**

**Answer:** JPEG: FF D8 FF E0
PDF: 25 50 44 46 (%PDF)
ZIP / DOCX: 50 4B 03 04 (PK..)
PNG: 89 50 4E 47

### Q10: How did you uncover the suspect's renamed file MEMO.TXT?

**Answer:** The suspect named it MEMO.TXT, but in HxD Hex Editor the first bytes were FF D8 FF E0. That proved it was actually a JPEG picture, not a text note!

### Q11: What happens under the hood when a file is deleted in FAT?

**Answer:** Windows does not wipe the actual contents. It only changes the first letter of the file's name in the directory table to 0xE5, and marks the clusters as available. The data stays on disk until overwritten.

### Q12: What is File Carving?

**Answer:** File carving is recovering deleted files directly from raw sectors and unallocated space by searching for file headers, cluster pointers, or sizes without needing the file table.

### Q13: How did you recover the deleted client list CLIENTS.CSV?

**Answer:** We looked at the directory table, found the entry marked with 0xE5, saw that its starting cluster was Cluster 5 (offset 0x4800), and read the 200 bytes directly from the raw sectors.

### Q14: What is Slack Space?

**Answer:** Slack space is the leftover empty space in a cluster between the end of a file's data and the end of the cluster. It often contains old residual data from previous files.

### Q15: Why is Chain of Custody so important in court?

**Answer:** Because if you cannot prove who held the evidence at every step, the defense lawyer can argue someone tampered with the device, and the judge will throw the case out.

## 5. Quick Numbers to Remember for Viva (Cheat Sheet)

| Concept / Fact | Exact Value / Answer to Say |
|---|---|
| **Case Number** | Case #DF-2026-INV12 |
| **Suspect Media Size** | 1,474,560 Bytes (1.41 MB / 2,880 Sectors) |
| **Sector Size** | 512 Bytes |
| **File System** | FAT12 (Volume Label: SUSPECT_USB) |
| **FAT Deletion Byte** | 0xE5 (Replaces first character in directory entry) |
| **Master SHA-256 Hash** | 3b5a7762f2c49c891da888c8be90f5c67baa4fdecda9c2764d94a8845e75a01c |
| **Avalanche Test Shift** | 127 of 256 bits changed (~49.6%) on 1-bit flip |
| **JPEG Magic Number** | FF D8 FF E0 (Start) / FF D9 (End) |
| **PDF Magic Number** | 25 50 44 46 (%PDF) |
| **Carved File Location** | Cluster 5, Sector 36, Byte Offset 0x4800 (200 Bytes) |
| **Forensic Standards** | ISO/IEC 27037 and NIST SP 800-86 |